

MedSim Lab

A glasses-free, camera-interactive 3D simulation platform for assessing medical students in university labs

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1. Executive Summary

MedSim Lab is a high-fidelity 3D simulation platform installed in dedicated university labs. Medical students explore realistic anatomical models on large screens, controlling them with hand gestures captured by an ordinary camera — no VR headset or special glasses required. The platform's primary purpose is **assessment**: students work through clinical cases and answer questions authored by academic partners, inside a visual context that resembles what they will face in practice.

Responsibilities are split cleanly. The **engineering team** owns the simulation, the 3D models and the platform. The **academic partners** own the theoretical content: cases, questions and answer keys. Content is rolled out **course by course**; each course ships as a complete package of models, cases and a question bank.

Key principle. Students are assessed on **theoretical knowledge**, never on how well they perform gestures. The camera is a navigation and exploration tool. It must never be the reason a student loses a mark.

2. Problem Statement

- Anatomy and pathology are learned largely from 2D atlases, slides and limited cadaver time. Students struggle to build an accurate 3D mental model of structures and their spatial relationships.
- Written exams test recall out of visual context. They do not reveal whether a student can recognize a structure or an abnormality when they actually see it.
- VR-based solutions are costly per seat, hard to manage with large groups, raise hygiene concerns with shared headsets, and can cause motion discomfort.
- Existing 3D anatomy products are mainly self-study tools, operated by mouse or touch, English-only, licensed per seat, and not aligned with local curricula or local assessment practice.

3. Product Vision and Positioning

Vision: every medical college lab becomes a place where students demonstrate understanding by seeing, exploring and identifying — not only by writing.

Dimension	Typical 3D anatomy products	MedSim Lab
Primary use	Self-study and teaching	Lab-based assessment against academically authored cases
Interaction	Mouse, touch, or VR headset	Hand gestures via a standard camera, with mouse/touch fallback
Depth perception	Flat rendering or headset	Head-tracked parallax on a large screen; glasses-free
Content	Generic global content	Course-by-course packages aligned with the partner college's curriculum
Authoring	Fixed vendor content	Academics author and version their own cases
Language	Mostly English	English medical terminology with bilingual (English/Arabic) interface

4. Goals and Non-Goals

Goals

- **G1.** Assess students' theoretical knowledge in a realistic 3D visual context.
- **G2.** Deliver a glasses-free experience that runs on lab PCs and large screens with a standard camera.
- **G3.** Let academic partners create and update cases without depending on developers for every change.
- **G4.** Ship content course by course, each course as a complete, academically validated package.
- **G5.** Give instructors actionable reports: which structures, concepts and cases students struggle with.

Non-Goals (current phases)

- Grading motor skills, dexterity or procedural technique.
- Surgical simulation with realistic soft-tissue deformation, cutting physics or haptic feedback.
- Use as the sole instrument for high-stakes final exam grades.
- VR/AR headsets, mobile apps, or students' personal devices.
- Replacing lectures, cadaver labs or clinical training.

5. Stakeholders and Responsibilities

Stakeholder	Owns	Key responsibilities
Engineering team	Simulation, 3D, platform	3D model production and optimization; rendering engine; camera interaction; case player; authoring tools; reporting; deployment and maintenance
Academic medical partners	Theoretical content	Course selection and sequencing; cases, questions and answer keys; anatomical and clinical accuracy sign-off; pilot evaluation
Partner university	Infrastructure	Dedicated labs, high-performance PCs, large displays, cameras, local server, network, scheduling
Instructors / proctors	Lab sessions	Running sessions, assigning cases, reviewing results
Students	—	Primary end users; feedback during pilots

The contract between teams is the **Case Template** (Section 10). Academics fill it in without technical knowledge; engineering builds the system that turns it into an interactive scene. Agreeing on this template early prevents endless back-and-forth change requests.

6. Users and Personas

Persona	Context	Needs	Success looks like
Student	Attends a scheduled lab session; one student per station	Learn the controls quickly; clear questions; no penalty for interaction glitches	Completes cases unaided after a short practice mode and trusts the result
Instructor / case author	Medical faculty member with limited technical background	Write cases in a familiar structure; choose the model, view and pathology; preview before publishing	Authors a new case in under an hour; sees where the cohort struggles
Proctor / lab admin	Runs sessions in the lab	Start sessions, assign cases, handle camera issues, lock stations	Sessions start on time; issues resolved without engineering support
Course coordinator	Owens a course's assessment plan	Approve case packages; compare cohorts over time	Validated package ready before the course starts

7. Core Design Principles

- **Knowledge, not dexterity.** Every answer can be given via a fallback input. Gesture failures never change scores.
- **One input language.** Gestures, mouse and touch produce the same semantic events (point, select, rotate, zoom). Case logic never depends on the input device.
- **Forgiving interaction.** Few, simple gestures; confirmation before submitting answers; easy undo and view reset.
- **Accuracy first.** No model or case reaches students without academic sign-off.
- **Privacy by design.** Camera frames are processed on the station and never stored or transmitted.
- **Content as data.** Cases are structured data, not code, so content scales without engineering bottlenecks.

8. User Journeys

Student session

1. Student signs in at a station (university ID). The station is locked to the application (kiosk mode).
2. Short camera calibration (about 10 seconds), then a 2–3 minute practice mode with a non-graded sample model.
3. Assigned case opens: clinical stem plus 3D model at a preset view. Labels are hidden in assessment mode.
4. Student explores (rotate, zoom, peel layers, isolate structures) and answers questions: selecting structures on the model or choosing options.
5. Each answer is confirmed before it is recorded. Student submits; a summary screen appears (feedback level configurable per case).

Case authoring and publishing

1. Academic fills in the Case Template: course, model, pathology variant, stem, questions, answer keys, points, feedback.
2. Engineering produces any missing model or pathology variant (tracked as a content request).
3. Author previews the case exactly as students will see it, then submits for review.
4. A second academic reviewer approves; the case is versioned and added to the course package.

Proctor session

- Create a session, select cohort and case set, start/stop timers, monitor station status (camera OK, in progress, submitted), and switch a station to fallback input when needed.

9. Functional Requirements

Priority: **P0** = required for MVP, **P1** = shortly after MVP, **P2** = later. The “Proto” column marks scope for the rapid prototype (Section 15).

9.1 3D viewer

ID	Requirement	Priority	Proto
V-01	Render high-fidelity anatomical models with realistic materials at interactive frame rates on lab PCs	P0	Yes
V-02	Rotate, zoom and pan around the model; reset to preset view	P0	Yes
V-03	Layer control: show/hide anatomical layers (e.g. pericardium, myocardium, vessels)	P0	Yes
V-04	Select, highlight and isolate an individual named structure	P0	Yes
V-05	Labels on/off; forced off in assessment mode	P0	Yes
V-06	Display pathology variants of a model (e.g. stenosed valve, infarcted region)	P0	1 variant
V-07	Clipping planes / cross-sections	P1	No
V-08	Physiological animation (e.g. cardiac cycle, blood flow)	P1	No
V-09	Comparison view: normal vs pathological side by side	P2	No

9.2 Camera interaction

ID	Requirement	Priority	Proto
I-01	Hand tracking from a standard webcam, processed locally in real time	P0	Yes
I-02	Core gesture set (table below), mapped to semantic input events	P0	Yes
I-03	Per-session calibration and a non-graded practice mode	P0	Yes
I-04	Confirmation step before an answer is recorded	P0	Yes
I-05	Fallback input (mouse, keyboard, touch) available at any time, switchable by proctor	P0	Yes
I-06	Head-tracked parallax for glasses-free depth perception	P1	Yes (experimental)
I-07	Visual feedback: hand cursor, tracking status indicator, “hand lost” warning	P0	Yes
I-08	Left- and right-handed support	P1	No

Initial gesture set (to be validated in the prototype):

Gesture	Action
Index finger movement	Move pointer
Pinch (thumb + index)	Select a structure or UI element
Pinch and drag	Rotate the model
Two-hand pinch, move apart / together	Zoom in / out
Open palm held ~1.5 s	Reset to the case's preset view
Pinch on on-screen controls	Layers, labels, next question, confirm answer

9.3 Case player

ID	Requirement	Priority	Proto
C-01	Load a case package (model, variant, initial view, questions) from structured data	P0	Yes (JSON)
C-02	Question type: identify a structure on the model	P0	Yes
C-03	Question type: multiple choice (single answer)	P0	Yes
C-04	Question types: multi-select and ordering/sequence	P1	No
C-05	Question type: short free text (manual grading)	P2	No
C-06	Automatic scoring against the author's answer key and points	P0	Yes
C-07	Per-case time limit and navigation between questions	P0	Basic
C-08	Configurable feedback: none, score only, or full explanations after submit	P1	Score only
C-09	Interaction log per attempt (answers, time per question, structures viewed)	P0	Local log

9.4 Case authoring

ID	Requirement	Priority	Proto
A-01	Structured case template (spreadsheet or form) converted to case data	P0	Spreadsheet
A-02	Web authoring form: pick model, variant, preset view, questions, answer keys	P0	No
A-03	Visual tagging: author clicks structures in 3D to set answer targets and camera views	P1	No
A-04	Preview as student	P0	No
A-05	Review and approval workflow (author → reviewer → published)	P0	No
A-06	Versioning: attempts are always linked to the exact case version used	P0	No
A-07	Bilingual fields for stem and feedback (English / Arabic)	P1	No

9.5 Reporting and administration

ID	Requirement	Priority	Proto
R-01	Per-student results per case and per course	P0	Summary screen
R-02	Per-question analytics: correct rate, average time, most common wrong selections	P0	No

ID	Requirement	Priority	Proto
R-03	Structure-level heatmap: which structures the cohort misidentifies most	P1	No
R-04	Cohort comparison across groups and years	P1	No
R-05	Export results (CSV / PDF)	P0	No
R-06	Item analysis (difficulty and discrimination indices)	P2	No
R-07	User, role, cohort and course management	P0	No
R-08	Session management and live station status for proctors	P0	No
R-09	Kiosk lockdown of stations during sessions	P0	Yes

10. Case Content Model

Content is organized as: **Course** → **Model (with variants)** → **Named structures** → **Cases** → **Questions**. Every structure in every model carries a stable ID mapped to standard anatomical terminology (Terminologia Anatomica), so answer keys stay valid when a model is improved or re-exported.

Case Template — fields filled by academics

Field	Description
Course / topic	Course the case belongs to and topic within it
Model and variant	Organ or region, and the normal or pathological variant to display
Initial view	Starting camera angle and visible layers
Clinical stem	Case scenario text shown to the student
Questions	Prompt, type, options, correct answer(s) as structure names or option keys, points
Feedback	Explanation shown after submission (if enabled)
Constraints	Time limit, labels on/off, allowed layers
Metadata	Author, reviewer, version, learning objectives, difficulty

Example case data (generated from the template)

```
{
  "case_id": "CARD-012",
  "course": "Anatomy II - Thorax",
  "model": "heart_v1",
  "variant": "mitral_stenosis",
  "stem": "A 45-year-old woman presents with exertional dyspnea and a history of rheumatic fever.",
  "initial_view": { "camera": "anterior", "hidden_layers": ["pericardium"] },
  "labels_visible": false,
  "time_limit_min": 10,
  "questions": [
    { "id": "q1", "type": "identify", "prompt": "Select the affected valve.",
      "answer": ["TA:mitral_valve"], "points": 2 },
    { "id": "q2", "type": "mcq", "prompt": "What is the most likely etiology?",
      "options": { "a": "Congenital", "b": "Rheumatic", "c": "Infective", "d": "Degenerative" },
      "answer": "b", "points": 1 }
  ],
  "meta": { "author": "...", "reviewer": "...", "version": 3, "status": "published" }
}
```

11. 3D Content Pipeline

3D production is the main bottleneck and the most specialized skill on the team. Each course requires its normal models **and** every pathology variant its cases call for, so variants must be planned into the course schedule.

Stage	Activity	Tools (indicative)
1. Source	Licensed high-quality anatomical models, or segmentation of real CT/MRI (DICOM) data	Licensed model libraries; 3D Slicer
2. Clean-up	Retopology, smoothing, separating each structure into a named mesh	Blender, ZBrush
3. Materials	Realistic PBR textures; consistent visual language across organs	Substance Painter
4. Variants	Pathological versions requested by cases	Blender, ZBrush
5. Optimize	glTF 2.0 export, Draco/Meshopt geometry compression, KTX2 textures, levels of detail	glTF-Transform, gltfpack
6. Tag	Assign stable structure IDs mapped to Terminologia Anatomica	Internal tool / naming convention
7. Validate	Academic review of accuracy; sign-off recorded per model version	Review checklist

Recommendation for speed: start the prototype and the first course from licensed models and adapt them; build models from DICOM data only where licensed models are missing or insufficiently accurate.

12. Technical Architecture

The platform is web-based, wrapped in a desktop shell for kiosk lockdown, and served from a local server inside the lab network. Web technology is well suited here because assessment is knowledge-based and does not require physics-heavy surgical simulation; it also unifies the student app, authoring portal and reporting in one stack.

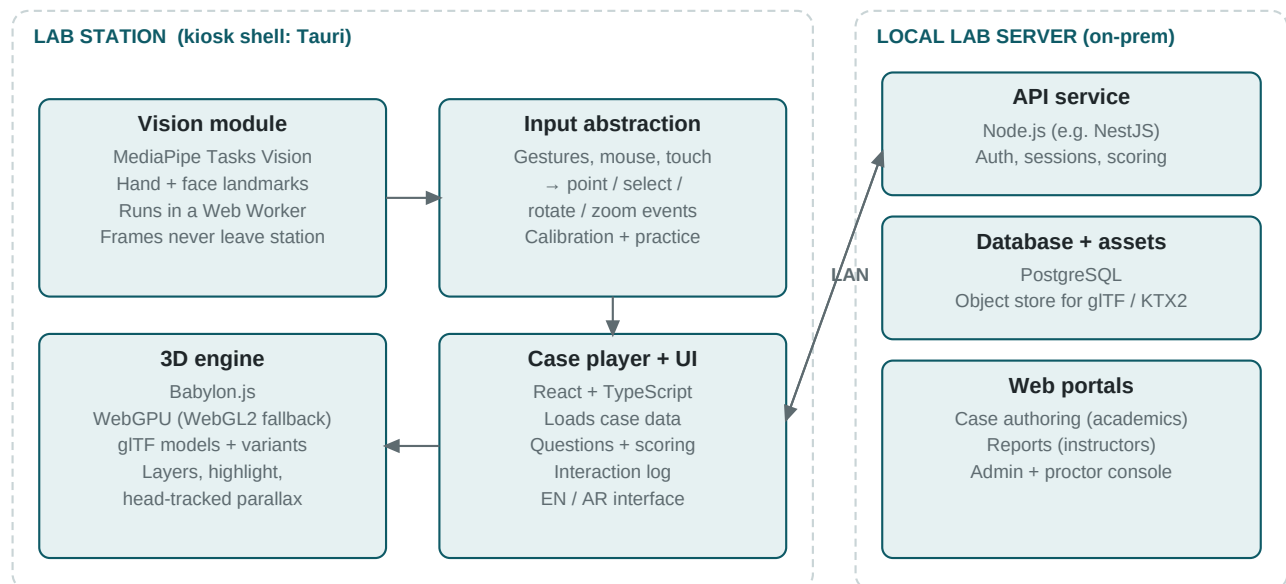


Figure 1. High-level architecture: station-side processing with a local lab server.

Layer	Recommended choice	Rationale
3D engine	Babylon.js with WebGPU, WebGL2 fallback	Complete engine (scene inspector, in-scene GUI, PBR, mature WebGPU support); fewer pieces to build than a lower-level library

Layer	Recommended choice	Rationale
UI	React + TypeScript	Shared component library across player, authoring and reporting
Vision	MediaPipe Tasks Vision (Hand Landmarker; Face Landmarker for head pose)	Official web support, runs fully on-device, no video upload
Desktop shell	Tauri (Electron as alternative) in kiosk mode	Locks the station to the app during sessions; small footprint
Backend	Node.js (e.g. NestJS) + PostgreSQL	Same language as the front end; relational data fits courses, cases, attempts
Assets	glTF 2.0 + Draco/Meshopt + KTX2, served over LAN	Fast loading of heavy models; caching at stations
Deployment	On-prem lab server; works without internet	Reliability during sessions; student data stays within the university

When to reconsider the engine: only if a later phase adds procedural or surgical simulation with realistic tissue deformation. At that point a native engine (Unreal or Unity) can be evaluated for that module alone.

13. Non-Functional Requirements

Area	Requirement
Rendering performance	≥ 60 fps at 1080p and ≥ 30 fps at 4K on the reference lab PC with the heaviest model of the current course
Interaction latency	Median hand-to-cursor latency under 100 ms
Load time	A case's model loads in under 5 seconds from the local server; cached loads in under 2 seconds
Reliability	A session continues if the server connection drops; attempts sync once it returns
Privacy	Camera frames processed on the station only; no video stored or transmitted; only landmarks used in memory
Security	Role-based access (student, author, reviewer, proctor, admin); kiosk lockdown; answer keys never sent to stations before submission is scored server-side (MVP)
Auditability	Every attempt linked to the exact case and model versions used
Accessibility	Adjustable text size; color-blind-safe highlights; fallback input always available
Localization	English medical terminology; English/Arabic interface with full RTL support
Maintainability	Content is data: new cases need no code release; new models need no engine changes

14. Lab Hardware and Environment

Component	Reference specification (to be confirmed by benchmark)
GPU	Discrete GPU, NVIDIA RTX 3060 / 4060 class or better, 8 GB+ VRAM
CPU / RAM	Modern 6+ core CPU; 16–32 GB RAM
Display	55–65" 4K screen per station
Camera	1080p at 60 fps, wide field of view, mounted top-center of the display

Component	Reference specification (to be confirmed by benchmark)
Lighting	Even front lighting; avoid windows or bright light behind the student
Network	Wired LAN to the local lab server
Future option	Glasses-free autostereoscopic (light-field) displays for true stereoscopic depth

15. Rapid Prototype (Phase 0)

Yes, a rapid prototype is feasible in about 6 weeks with a small team, provided it starts from a licensed high-quality model rather than building one from medical imaging data. Its purpose is to prove the three riskiest assumptions: rendering quality in the browser, reliability of camera interaction, and that academics' cases can drive the system as data.

Scope

- **One organ** (recommended: the heart) with separated, named structures and **one pathology variant**.
- **Three cases** with about 10 questions in total, authored by academic partners in a spreadsheet template and converted to JSON.
- Question types: identify-on-model and multiple choice, with automatic scoring and a results summary.
- Hand-gesture control (core gesture set), calibration, practice mode, answer confirmation and mouse fallback.
- Head-tracked parallax as an experimental toggle, to evaluate the glasses-free depth effect.
- Runs on one lab station in kiosk mode; no backend — cases and logs stored as local files.

Out of scope for the prototype

- Authoring portal, approval workflow, user accounts, analytics dashboards, cross-sections, animations, multiple courses.

Plan

Week	Deliverable
1	Technical spike: licensed heart model rendered in Babylon.js (WebGPU); MediaPipe hand tracking running in the browser; first fps and latency measurements on a lab PC
2	Gesture layer mapped to semantic input events; calibration screen; mouse fallback; head-tracked parallax experiment
3	Model split into named structures; selection, highlighting, layer control; first pathology variant
4	Case player reading JSON: stem, identify-on-model and MCQ questions, confirmation, scoring, summary; practice mode
5	Kiosk shell and installation on a lab station; academics author three cases via the spreadsheet template
6	Pilot with 10–20 students; usability questionnaire; review interaction logs; go / no-go decision

Prototype team

- 1–2 web/3D developers (Babylon.js, TypeScript)
- 1 3D artist, part-time (model clean-up, structure separation, pathology variant)
- 1 academic lead (cases, accuracy sign-off, pilot coordination)

Exit criteria

Criterion	Target
Rendering	≥ 60 fps at 1080p on the lab PC with the heart model
Selection reliability	≥ 95% of intended pinch-selections succeed within two attempts
Latency	Median hand-to-cursor latency under 100 ms
Learnability	≥ 80% of pilot students complete a case without proctor help after practice mode
Usability	System Usability Scale (SUS) score ≥ 68
Accuracy	Academic lead signs off the model and all three cases

16. Roadmap

Phase	Timing (indicative)	Scope
0 — Prototype	~6 weeks	One organ, three cases, gestures, JSON cases, single station
1 — MVP: first course	One semester	All models for the first course; backend and accounts; authoring form; review workflow; basic reports; formative (non-graded) use across the lab
2 — Authoring & analytics	Following semester	Visual 3D authoring; structure heatmaps and item analysis; cross-sections and animations; second course
3 — Scale	Year 2+	Course-by-course expansion; other colleges and universities; multi-tenant deployment; optional autostereoscopic displays

17. Success Metrics

Metric	Why it matters
Case completion rate without proctor help	Interaction is learnable and does not block assessment
Gesture error rate and fallback usage	Camera reliability in real lab conditions
Average time to author and publish a case	Content can scale without engineering bottlenecks
Instructor adoption (active authors and sessions per course)	Faculty see value and keep using the platform
Student satisfaction (SUS and perceived usefulness)	Experience quality
Correlation between platform scores and written exam scores	Evidence that the platform measures real knowledge; basis for joint publications

18. Risks and Mitigations

Risk	Impact	Mitigation
Scarcity of 3D artists with medical modeling experience	High	Start from licensed models; recruit early; build a reusable naming and validation pipeline
Anatomical inaccuracy in models or variants	High	Mandatory academic sign-off per model version; reference imaging data
Camera tracking unreliable (lighting, distance, backgrounds)	Medium	Lighting guidelines; calibration; forgiving gestures; always-available fallback input

Risk	Impact	Mitigation
Pathology variant workload grows faster than the team	Medium	Plan variants with each course package; prioritize by number of cases using them
Browser performance limits with heavy models	Medium	Compression, LODs, benchmarks in week 1; native engine kept as an option for special modules
Scope creep toward surgical simulation	Medium	Explicit non-goals; separate future track if ever pursued
Academic content throughput slower than engineering	Medium	Simple templates; authoring tool in Phase 2; agreed calendar per course
Student data privacy concerns over camera use	Low–Medium	On-device processing only; clear signage and consent; no video stored

19. Open Questions

- Which course launches first, and which organs and pathology variants does it require?
- Interface and content language: English terminology with an Arabic interface, or fully bilingual content?
- Will results carry any grade weight, or remain formative only in Phase 1?
- One student per station, or pairs? How many stations per lab?
- Integration with the university's LMS or student information system (accounts, cohorts, results)?
- Budget and licensing approach for 3D models (licensed libraries vs. in-house production from imaging data).
- Intellectual property and publication rights between the engineering team, the university and the academic partners.
- Final product name and branding.

20. Glossary

Term	Meaning
Case	A scenario with a model, view, clinical stem and questions, authored by academics
Course package	All models, variants, cases and questions for one course, released together
Pathology variant	A modified version of a normal model showing a disease or abnormality
Head-tracked parallax	Rendering technique that shifts the 3D view with the viewer's head position, creating depth on a flat screen
Kiosk mode	Station locked to a single application during sessions
glTF	Standard file format for delivering 3D models efficiently on the web
WebGPU	Modern browser graphics API offering near-native GPU performance
Terminologia Anatomica	International standard for anatomical terminology, used for structure IDs
SUS	System Usability Scale, a standard 10-item usability questionnaire